
DIGITAL TWIN

**We Built a Virtual Machine That Never
Breaks - But Tells You When Yours Will!
...And Why!**

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Yaxin

- Data Scientist, Lithium-Ion Battery, Process Quality Control
- [linkedin.com/in/tangyaxin](https://www.linkedin.com/in/tangyaxin)



Diane

- Data processing, analysis, Machine Learning with first experience in agra sector
- [linkedin.com/in/diane-djuigne-tuma](https://www.linkedin.com/in/diane-djuigne-tuma)



Gunar

- Data expert with strong background in the energy sector
- [linkedin.com/in/gunar-kachel](https://www.linkedin.com/in/gunar-kachel)



Vinayak

- Project Management, Test Management, Data Scientist, Automotive
- [linkedin.com/in/vinayak-betageri](https://www.linkedin.com/in/vinayak-betageri)



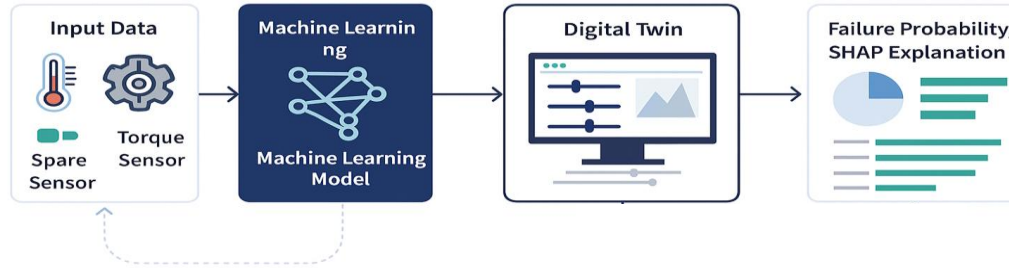
Project Definition

Problem:

- Maintenance is important and includes three types: **preventive, reactive, and predictive.**
- The **Reactive approach** (“fix it when it breaks”) leads to high unplanned downtime and excessive tool wear.
- Digital twins optimize maintenance and machine settings.
- But only **29% of manufacturing companies** worldwide Digital Twin



Virtual Machine



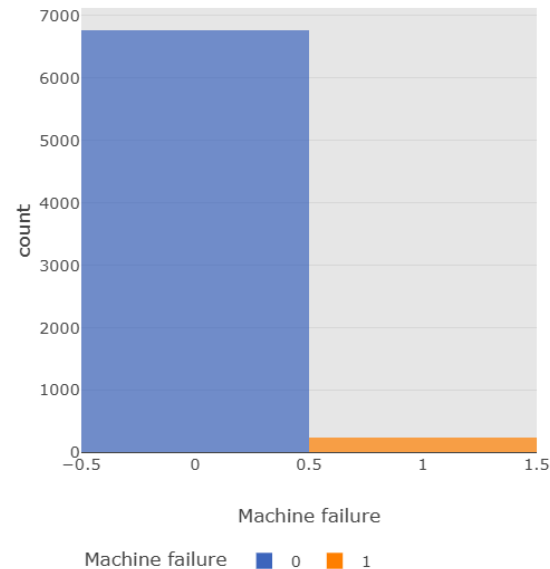
What Virtual Machine does?

- Predicts machine failures to define correct operating conditions.
- Show which feature is influencing the failure prediction
- Allow engineers to test scenarios → **“What if we increase the load?”**

Data Overview

Sensor data of a industrial milling machine from a published paper of AI for Industries

Observation	Features	Target
• 10k • 7k training • 3k testing	• Type (product) • Air temperature • Process temperature • Rotational speed • Torque • Tool wear (time)	• Machine failure



- **Strong imbalance in Target**
- **To deal with it we used class weights**

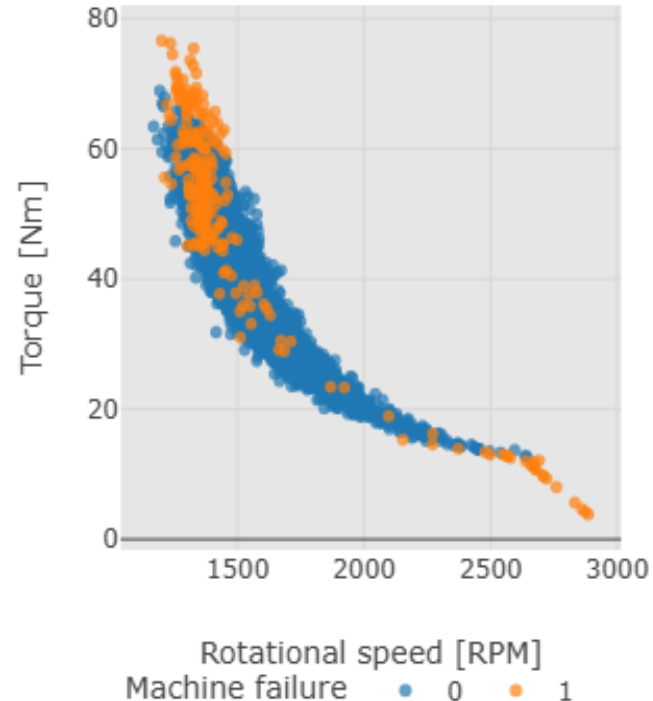
Feature Engineering & Key Insights

Engineered Feature

- Power
- Temperature Difference
- Omega (Angular Velocity)
- ...

Insights

- Failures strongly linked to **high tool wear**, **high torque**, and **rotation speed**
- Engineered features significantly improved predictive signal



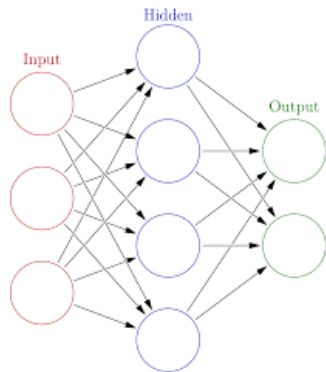
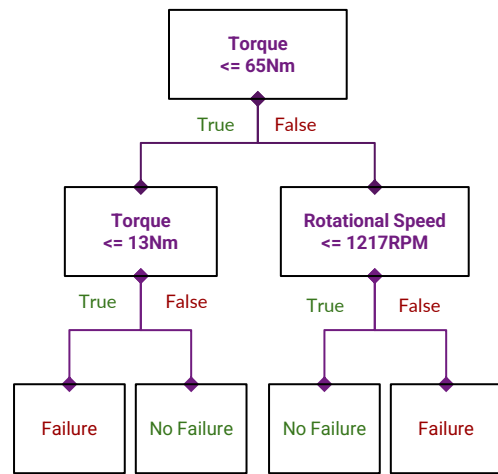
Modeling Strategy

Tree Based Models

- **Baseline - Shallow Decision Tree (depth = 2)**
Simple, rule-based reference model as baseline
- **Decision Tree (unrestricted)**
A single tree fully fitting data patterns
- **Random Forest**
Multiple independent decision trees
- **XGBoost**
Trees are built sequentially to correct previous errors

Neural Network

- **Multi-Layer Perceptron (MLP)**
Included as a non-tree-based benchmark



Results

Evaluation Matrix - Failure-focused

We want the model to:

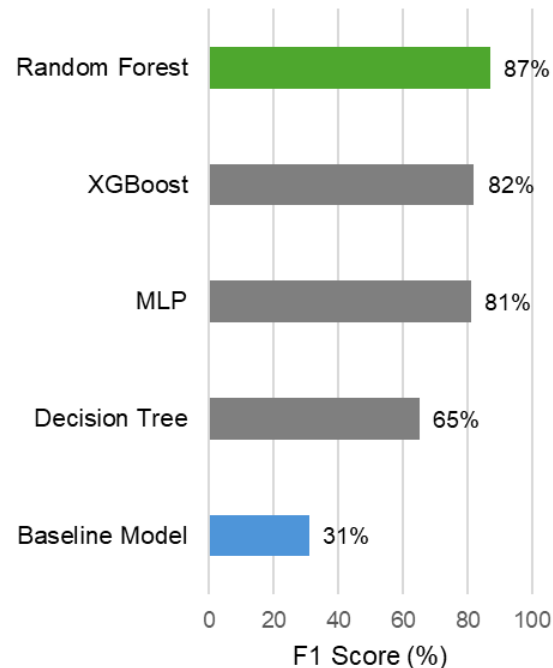
1. Maximizing identification of real failures (recall)
2. Minimizing wrong warning (precision)

Solution: F1 score as it provides a balance between recall and precision.

Model Selection

- Baseline model: limited performance (F1 = 31%) => reference only
- Random forest: 87% => best F1 score among tested models

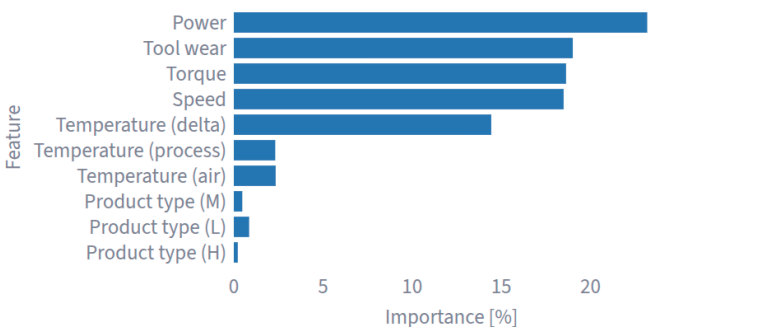
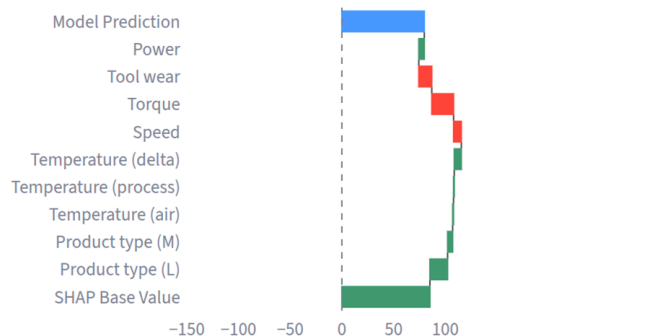
**Model Comparison - F1 Score
for Failure Class**



Prediction Explanation

Why was a machine failure predicted?

- Reasons for prediction are often seen as black-box.
- However, there are tools to understand decision mechanism, for example...

(General) Feature Importance of the <u>model</u>	Shapley additive explanations (SHAP) of a <u>single prediction</u>																																																
Random Forest: Mean decrease in impurity (MDI) in the model by a specific feature	Added influence of a specific feature in a set of inputs to a single prediction																																																
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Product: Digital Twin Dashboard

Feature Importance for Prediction

For a specific prediction the importance of a feature depends on its value and the interaction with other features. Below shapley additive explanations (SHAP) for the prediction of selected machine settings.

Input Machine Settings

Process temperature [K]



Air temperature [K]



Product quality



Rotational speed [RPM]



Torque [Nm]



Tool usage [min]

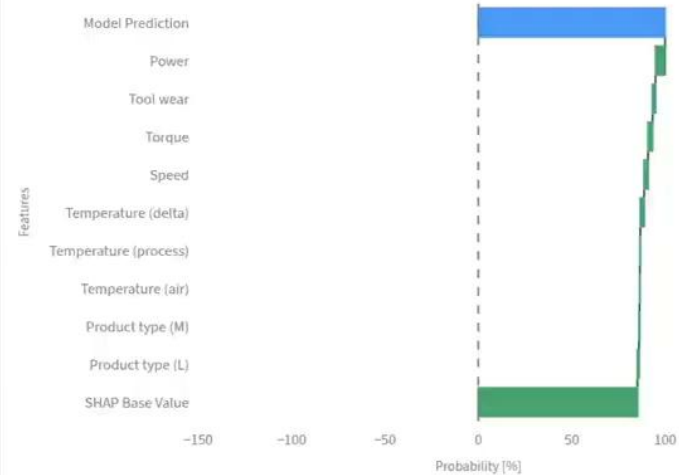


change settings

Predicted Results

With probability of 100% these settings will result in:

✓ Safe Operations

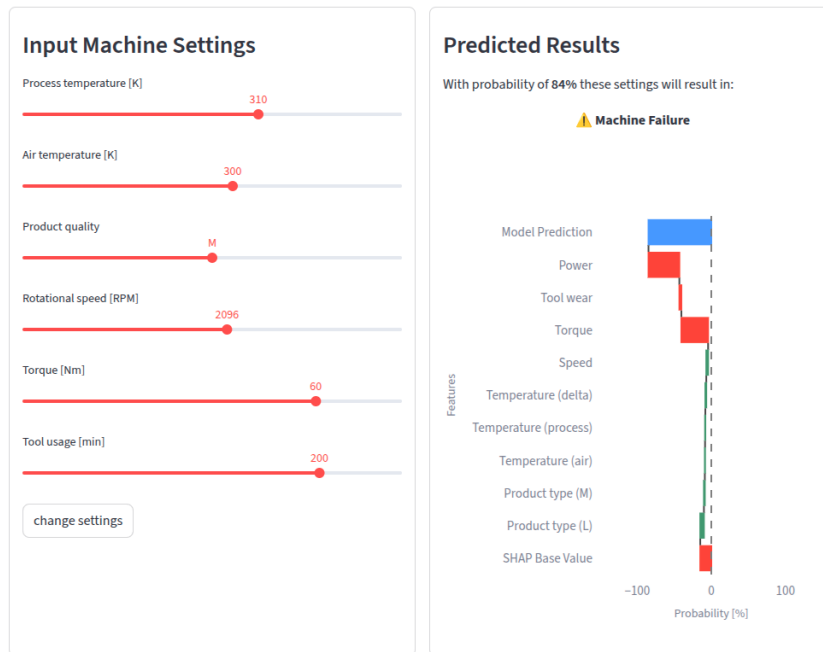


Conclusion

- Our virtual machine predicts potential failures early by simulating real operating conditions.
- The Random Forest model delivered the most accurate results
- SHAP clearly highlighted the key factors influencing failure prediction.
- Overall, the system simplifies planning, reduces downtime, and enhances efficiency.

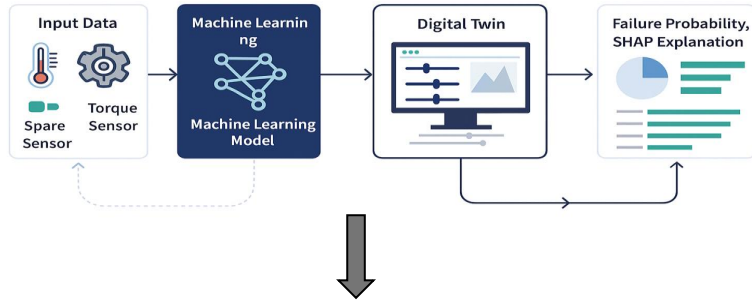


Digital Twin



Product Increments

MVP - Minimum Viable Product

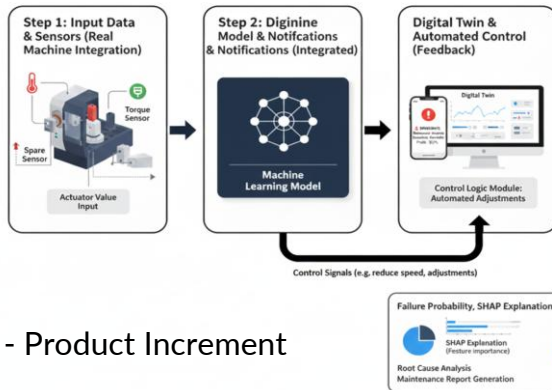


MVP - A virtual simulation where operating conditions can be tested and machine parameters adjusted digitally.

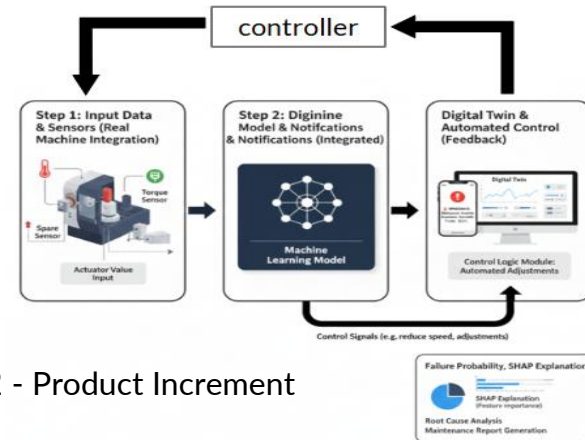
PI1 - Integration of real sensor data with automatic alerts when thresholds are exceeded.

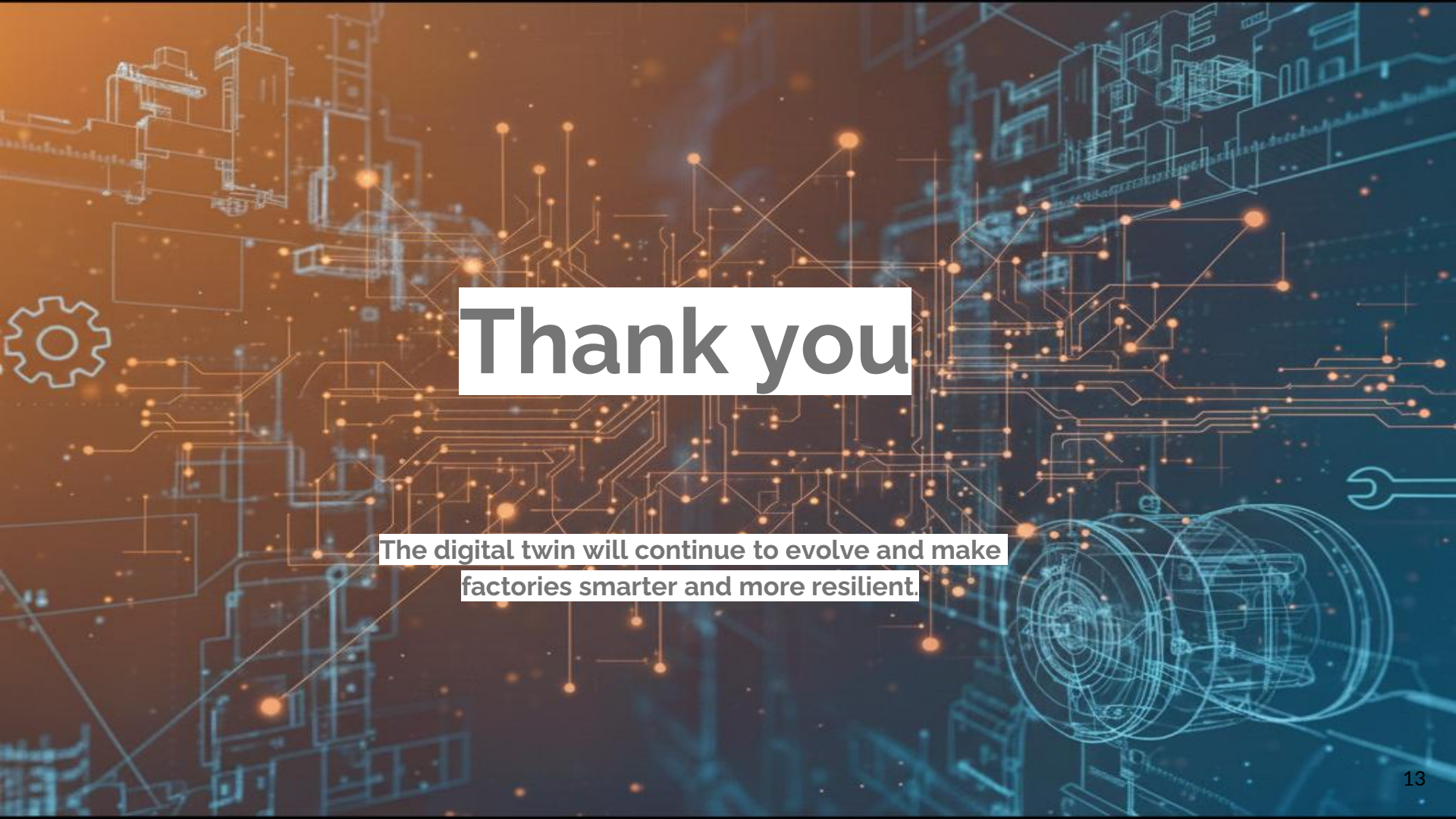
PI2 - Automated machine adjustments based on real-time Digital Twin feedback.

PI 1 - Product Increment



PI 2 - Product Increment





Thank you

The digital twin will continue to evolve and make factories smarter and more resilient.